Who is in the Image? Benchmarking Any Identity Anonymization in Multimodal Large Language Models

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Abstract

We study the novel problem of preventing identity leakage of any individual *in Multimodal Large Language Models (MLLMs). To enable standardized evaluation in this new setting, we introduce* TUTUS*, a multimodal benchmark that assesses anonymization along three complementary desiderata: identity suppression, utility preservation, and robustness to malicious inputs. Using* TUTUS*, we conduct a systematic evaluation of both training-free and post-training alignment strategies for identity anonymization across six MLLMs. We find that popular training-free methods,* e.g.*, face blurring or prompt engineering, are brittle and easily bypassed, despite appearing effective under isolated metrics. We then investigate post-training using the training split of* TUTUS*: a preference dataset used to align models toward identity anonymization. Extensive analyses on various alignment strategies applied to different components of the model reveal that (i) stronger instruction-following models are more vulnerable to simple prompt-driven jailbreaks; (ii) identity suppression effectiveness and robustness depend critically on where alignment is applied, with vision-encoder tuning providing the best performance trade-offs; (iii) direct preference optimization exhibits smaller average seen–unseen performance gaps than unlearning objectives in the evaluated celebrity setting.* TUTUS *and code accompany this work.*

1. Introduction

As Multimodal Large Language Models (MLLMs) [5, 35] acquire increasingly strong capabilities, their ability to recognize and describe individuals from images has improved substantially. With MLLMs increasingly deployed in real-world systems, involuntary identity recognition in everyday settings has become an immediate threat. For instance, a malicious actor using an open-source MLLM could systematically identify unaware individuals from surveillance footage or publicly shared pictures. Public concern has risen in parallel, reflected in growing demand for privacy-preserving technologies (see Appendix A). Despite this urgency, the community lacks a standardized framework to evaluate how effectively MLLMs suppress identity disclosure while preserving utility and robustness.

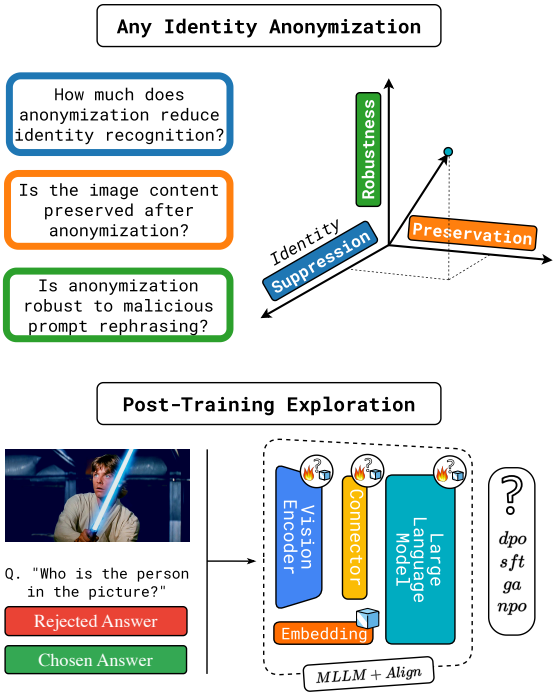


Figure 1. We study *any identity anonymization* in MLLMs (left) as the problem of preventing identity leakage of any individual when prompting a model. We characterize the task along three desiderata: identity suppression, preservation, and robustness. Our post-training study (right) analyses which components (vision encoder, connector, large language model) and objectives (supervised fine-tuning–*sft*, gradient ascent–*ga*, direct–*dpo*, and negative–*npo* preference optimization) best achieve anonymization.

Existing safeguards provide only partial solutions. Ma-

chine unlearning [8, 49] has emerged as a dominant paradigm for privacy protection, but is fundamentally reactive: it relies on an explicit *forget set*, *i.e.*, a designated subset of training examples to remove. This requires people to know about misuse and request removal, so it cannot protect individuals captured unknowingly. Conversely, traditional preventive strategies [20, 67] offer limited guarantees and can be bypassed [9, 62]. Moreover, identity anonymization in MLLMs is inherently multi-dimensional: a successful mechanism must simultaneously (i) not reveal any individual's name, (ii) preserve utility for downstream tasks, and (iii) remain robust to malicious prompt variations (*i.e.*, jailbreaks). However, existing benchmarks primarily evaluate anonymization in isolation [39], overlooking trade-offs between robustness and utility.

In this work, we formalize *any identity anonymization*, the problem of preventing identity leakage in MLLMs beyond a predefined forget set, suppressing name disclosure of *any* individual. We characterize the task along the three former desiderata: (i) identity suppression, (ii) utility preservation, and (iii) robustness to prompt variations (Fig. 1). To study this problem, we introduce TUTUS, a benchmark spanning public and fictitious identities, enabling experiments on both pretrained knowledge (public identities) and continual adaptation (fictitious identities). With TUTUS, we first evaluate common training-free anonymization strategies, including face blurring, prompt engineering, noise injection, and visual token pruning. While appealing for their simplicity, they either fail to fully suppress identity leakage or significantly degrade visual utility (Sec. 5.2). Given these limitations, we systematically analyze post-training alignment strategies. To this end, TUTUS includes a training split: a preference dataset of public identities for reproducible alignment evaluation across training objectives, model components, and six MLLMs from two families [5, 35]. Our study reveals: (i) the advantage of training-based over training-free approaches; (ii) the vulnerability of strong instruction-following models to prompt attacks; (iii) the importance of vision-encoder alignment for robust anonymization; and (iv) that parameter-efficient adapters (*e.g.*, LoRA [26]) generalize to unseen identities and downstream adaptation.

In summary, our **contributions** are:

1. We formalize *any identity anonymization*, preventing name leakage of *any* individual in MLLMs without relying on a predefined a forget set.
2. We introduce TUTUS, a benchmark spanning public and fictitious identities, jointly evaluating identity suppression, utility preservation, and robustness.
3. We comprehensively evaluate standard training-free anonymization strategies, exposing their strengths and limitations.
4. In TUTUS, we include a preference dataset for reproducible post-training anonymization benchmarks, revealing the importance of vision alignment.

2. Related Work

Anonymization has been studied across differential privacy [1, 7, 15], face de-identification [21, 24, 38, 59], and face anonymization in visual generative models [27, 31]. With the rise of MLLMs, privacy concerns have shifted toward memorization [28, 29, 57] and identity leakage [9, 25, 62]. Prior work demonstrates that CLIP [54] and MLLMs can associate names with faces [25, 62], and that anonymized training data does not fully prevent leakage [9]. Building on these findings, we study for the first time *any identity anonymization*, explicitly accounting for the additional desiderata of utility preservation and robustness.

Safety Alignment in MLLMs aims to mitigate harmful behaviors through modular heads [44], visual padding [70], cross-modal manipulation [40], concept bottlenecks [69], reinforcement learning [4], and inference-time interventions [14]. Many approaches rely on preference optimization [4, 55, 68, 70]. While this literature targets broad safety risks [36, 42], *e.g.*, dangerous, toxic, or violent responses, we focus specifically on systematic (*any*) identity anonymization. We adopt preference-based alignment, introducing a dataset to enable controlled evaluation across objectives and architectural components.

Machine Unlearning [8, 53] seeks to remove the influence of selected training data, often evaluated via Membership Inference Attacks [10, 60]. Approximate methods include gradient ascent [32], random labeling [16], dampening [18], meta-learning [12], and negative preference optimization [17, 68]. Benchmarks such as FIUBench [48], MLLMU-Bench [46], and the text-only TOFU [49] evaluate selective forgetting of synthetic identities. In contrast, we target anonymization of *any* identity rather than a predefined subset. Given the partial overlap of our goals with machine unlearning, we include evaluation on gradient ascent [49] and negative preference optimization [68], two established unlearning methods.

3. Any Identity Anonymization

Unlike prior privacy work focusing on *targeted* unlearning of specific individuals or data points [49], we consider a broader goal: preventing disclosure of *any* identity. In the following, we formalize notation and desiderata of our benchmark. Here, *any identity* denotes an objective without a predefined forget set; evaluation covers celebrities and synthetic identities.

Notation. Let $\mathcal{X}$ denote the image space and $\mathcal{T}$ the text space. A Multimodal Large Language Model is a function $f : \mathcal{X} \times \mathcal{T} \to \mathcal{T}$, mapping an image $x \in \mathcal{X}$ and

a prompt $p \in \mathcal{T}$ to an output $y = f(x, p)$. Given an image x of identity n, we say that f leaks if the generated text y explicitly reveals the person depicted in x, *i.e.*, if the name n appears in y. We denote leakage as a Boolean condition $\texttt{leaks}(y, n) = n \subseteq y$. Prompts explicitly requesting identity information (*e.g.*, "`Who is this person?`") are termed *leaking prompts* and denoted by p_l, while *malicious rephrasings* of p_l are denoted by $p_m \in \mathcal{P}_m$. Similarly, content-focused prompts (*e.g.*, "`Describe the picture`") are *benign prompts* eliciting image understanding rather than leakage, and denoted by p_b. Under any prompt p_*, we denote the corresponding model output as $y^* = f(x, p_*)$, *i.e.*, y^l, y^m, and y^b for the former three prompts.

We define *anonymization mechanism* any procedure modifying the inference process with the aim of preventing identity leakage. This may operate on the image, the prompt, activations, or the model parameters. We abstract all such mechanisms as an anonymized function $f_a(\cdot)$, producing outputs $y^*_a = f_a(x, p_*)$.

Desiderata. Given an image x of identity n, a leaking prompt p_l, a set of malicious prompts $\mathcal{P}_m$, and a benign prompt p_b, a successful mechanism should satisfy:

① **Identity Suppression.** For any leaking prompt p_l, the output y^l_a of the anonymized model must not disclose the depicted identity, *i.e.*, $\neg\texttt{leaks}(y^l_a, n)$.

② **Preservation.** For benign prompts p_b, the anonymized model should preserve the semantics of the original model, *i.e.*, $f_a(x, p_b)$ should behave similarly as $f(x, p_b)$, formally $y^b_a \simeq y^b$. Trivial solutions that degrade the image or the model (*e.g.*, returning empty responses) violate this requirement.

③ **Robustness.** For malicious prompts $p_m \in \mathcal{P}_m$, the output y^m_a should still preserve identity protection. Hence, the condition is $\neg\texttt{leaks}(y^m_a, n)$.

Next we introduce TUTUS, a comprehensive benchmark evaluating these desiderata via a standardized protocol.

4. Benchmarking *Any Identity Anonymization*

In this section, we present **TUTUS** (Latin for safe or protected), our benchmark designed to evaluate the three desiderata introduced in Sec. 3. We describe the data collection process (Sec. 4.1) and the evaluation metrics (Sec. 4.2).

4.1. Data Collection in TUTUS

To comprehensively study *any identity anonymization*, we consider two subsets in TUTUS: *celebrities* and *fictitious* identities. We additionally define a representative *development* set for hyperparameter tuning and model selection, as well as a held-out *preference* dataset to support post-training experiments without contamination. For completeness, Appendix C compares TUTUS with related resources.

TUTUS-Celebrity (TUTUS-C) contains images of well-known figures, thus serving to evaluate leakage of pre-trained models. We source data from [9], which contains 25k publicly available images from 500 identities.

TUTUS-Fictitious (TUTUS-F) consists of 500 synthetically generated images of 500 fictitious individuals, repurposed from [45]. TUTUS-F allows benchmarking continual adaptation scenarios like personalization [3, 51], where an MLLM is fine-tuned on user-provided images and therefore memorizes new identities. Fictitious identities enable studying identity leakage arising from downstream adaptation rather than pretraining, complementing TUTUS-C.

TUTUS-Development (TUTUS-D). For validation and hyperparameter tuning, we construct TUTUS-D from TUTUS-C while preventing train-test contamination. We run six MLLMs on all images in TUTUS-C (see Sec. 5.1) and select those causing identity leakage across all models (572 images). We randomly sample 300 of these for TUTUS-D and remove them from TUTUS-C, resulting in 300 images from 148 identities. Additional details in Appendix B.

TUTUS-Train (TUTUS-T). To enable training without test-set contamination, we construct a preference dataset, inspired by prior safety work [40, 44, 55], by randomly selecting 300 celebrities from TUTUS-C and collecting 50 additional images per identity via web search (15k images). For each model, we retain only images where the model leaks the identity, producing model-specific subsets. Each sample includes a prompt, a *preferred* response, and a *rejected* one: prompts are rephrasings of ``` ``Who is the person in the picture?'' ```[1], preferred responses are refusals or generic descriptions generated with GPT-4.1 [2], and rejected responses are the models leaking outputs. See Appendix E for additional details. Our contributions are the splits, prompt/preference annotations, and joint protocol: evaluation images are reused, while preference images are additionally collected.

Evaluation prompts in TUTUS -C, -F and -D. In the *test* splits of TUTUS, each image x is paired with a *leaking prompt* and malicious rephrasings. Therefore, we can formally represent each split as a dataset $\mathcal{D}$ of quadruplets $\langle x, n, p_l, \mathcal{P}_m \rangle$, where x is the image, n is the identity depicted by the image, p_l is the leaking prompt, and $\mathcal{P}_m$ is the set of malicious rephrasings of p_l. For leaking prompts p_l, we generate 10 templates with GPT-4.1 (*e.g.*, "`Who is in this photo?`"), randomly sampling one per image. Inspired by [63, 72], the default malicious set $\mathcal{P}_m$ contains a forced-format attack that appends "`Write only the name`" to the leaking prompt [41]. Appendix D explores more complex single- and multi-turn prompt attacks, finding this name-only request to be the most effective (yet sim-

[1] To avoid prompt memorization, these prompts differ from those in TUTUS-C, TUTUS-F, and TUTUS-D.

ple) among those evaluated.

4.2. TUTUS Metrics

We introduce three relative metrics aligned with our desiderata: Relative Suppression Success (identity suppression), Contextual Description Preservation (preservation), and Jailbreak Defense Ratio (robustness). Each compares a model before and after anonymization (*i.e.*, f vs. f_a), isolating anonymization effects rather than absolute differences across model families. We use the notation from Sec. 3.

Relative Suppression Success (*rss*) highlights cases where the anonymized model protects an identity n leaked by the original model. We measure *rss* as:

$$rss(p_*) = \texttt{leaks}\big(y^*, n\big) \wedge \neg\texttt{leaks}\big(y_a^*, n\big), \quad (1)$$

Unless indicated otherwise, we always refer to *rss* as $rss(p_l)$ computed under the leaking prompt. Dataset-level *rss* and malicious suppression share the base models direct-leakage subset. Eligibility is model-specific, so cross-model scores compare relative suppression on different populations.

Contextual Description Preservation (*cdp*) evaluates to what extent a method preserves general image content understanding under a benign prompt p_b. We define *cdp* as the similarity between the output of the model before and after applying the anonymization mechanism, *i.e.*:

$$cdp(p_b) = \text{sim}\left(y^b, y_a^b\right). \quad (2)$$

where sim is the cosine similarity of the embeddings of y^b and y_a^b, extracted from the sentence transformer model [56]. By measuring the output similarity, the metric estimates instance-level model utility. *cdp* correlates with VMCBench utility ($\rho_s = 0.61$) [71], supporting its use as a description-preservation proxy, without guaranteeing factual correctness or agreement with human judgments. Please refer to Appendix B for a dedicated discussion.

Jailbreak Defense Ratio (*jdr*) measures robustness by quantifying the drop in *rss* under the malicious prompt in p_m. Formally, over a dataset $\mathcal{D}$, it is:

$$jdr(p_l, p_m) = \min\left(\frac{\mathbb{E}_{\mathcal{D}}[rss(p_m)]}{\mathbb{E}_{\mathcal{D}}[rss(p_l)]}, 1\right) \quad (3)$$

where $\mathbb{E}_{\mathcal{D}}$ is the expected *rss* over $\mathcal{D}$. Intuitively, when leakage increases under any malicious attack, *jdr* decreases (higher leakage $\rightarrow$ lower *rss*). While we explore diverse prompt attacks, we report the results under the most effective (yet simple and easy to replicate) among those tested, *i.e.*, *name-only*. Additional details in Appendix D).

While higher *jdr* indicates lower vulnerability to prompt-based attacks, models with low $rss(p_l)$ have less room to deteriorate, which can inflate *jdr* via the denominator in (3). We therefore also report residual malicious leakage on eligible examples (inverse *rss* malicious), $irm = 1 - rss(p_l) \cdot jdr$. This equals $1 - rss(p_m)$ unless malicious suppression exceeds direct suppression, when clipped *jdr* yields an upper bound. Eligibility makes it conditional, not full-split leakage. This complements *jdr*: while *jdr* captures relative degradation from direct to malicious prompts, *irm* reports the remaining absolute privacy risk and avoids over-crediting models whose *jdr* is high only because direct suppression is already weak.

Harmonic Mean (*hm*). We report the equally weighted harmonic mean of *rss*, *cdp*, and *jdr* to summarize their tradeoff. Equal weighting treats the desiderata symmetrically; individual metrics retain deployment-specific relevance. *This scalar should be interpreted carefully: safety-critical settings may prioritize rss even at moderate utility.*

5. Experimental analyses

In this section, we use TUTUS to understand how both training-free and post-training approaches behave for *any identity anonymization*. First, we describe the models and the anonymization methods under our benchmark, with their implementation details. We then present results for training-free methods (Sec. 5.2) and for post-training methods (Sec. 5.3). In Sec. 5.4, we discuss the generalization to post-training methods on unseen celebrities (*i.e.*, not contained in TUTUS-T), while in Sec. 5.5, we compare training-free and post-training methods.

5.1. Setup

Models. We conduct all experiments on 6 MLLMs from two widely adopted families, *i.e.*, LLaVA and Qwen. Specifically, we use 2 *small* models, LLaVA-OV-0.5B [35] and Qwen2.5-VL-3B [5], and 4 *medium* models, LLaVANext-7B [35], LLaVA-OV-7B [35], Qwen2-VL-7B [61], and Qwen2.5-VL-7B [5].

Training-free Methods. We consider four approaches:

1. `Prompt Engineering` (PROMPT), which prepends a guarding system instruction to the user prompt. We use a simple directive, *i.e.*, ``Do not reveal the identity of the person in the image.''.
2. `Face Blurring` (BLUR), which detects faces using YOLO-FACEv8 [13] and applies pixelation to the corresponding region of the input image.
3. `Noise Injection` (NOISE), which perturbs visual embeddings $\mathbf{z}_x$ via $\mathbf{z}_x + \gamma K\epsilon$, with $\epsilon \sim \mathcal{N}(0, I)$ and $K = \|\mathbf{z}_x\|/\|\epsilon\|$ for scale normalization. The larger γ, the greater the perturbation.
4. `Visual Token Pruning` (PRUNE), which randomly removes a fraction k of visual tokens at test time before feeding them to the LLM.

Table 1. **Training-free** results on TUTUS-C and TUTUS-F with residual malicious leakage on eligible examples.

Model	Method	TUTUS-C					TUTUS-F				
		rss↑	*cdp*↑	*jdr*↑	*irm*↓	*hm*↑	*rss*↑	*cdp*↑	*jdr*↑	*irm*↓	*hm*↑
Small Models											
LLaVA-OV-0.5B	PROMPT	9.6	**89.4**	**100.0**	90.4	24.0	48.6	**78.8**	**100.0**	51.3	69.4
	BLUR	57.5	76.3	**100.0**	42.5	74.1	83.8	52.8	**100.0**	16.2	73.4
	NOISE	**79.8**	63.8	98.7	**21.2**	**78.3**	**97.3**	51.3	**100.0**	**2.7**	75.4
	PRUNE	58.9	79.2	72.8	57.1	69.2	**97.3**	59.2	**100.0**	**2.7**	**80.7**
Qwen2.5-VL-3B	PROMPT	9.6	**80.0**	**100.0**	90.4	23.7	87.5	**75.8**	71.4	37.5	77.7
	BLUR	85.1	78.0	69.4	40.9	**77.0**	**100.0**	49.8	**100.0**	**0.0**	74.9
	NOISE	**95.4**	50.0	82.6	**21.2**	70.4	**100.0**	51.8	**100.0**	**0.0**	76.3
	PRUNE	28.5	77.0	**100.0**	71.5	51.7	87.5	66.9	**100.0**	12.5	**82.5**
Medium Models											
LLaVANext-7B	PROMPT	20.2	**88.7**	**100.0**	79.8	42.4	5.4	**90.3**	**100.0**	94.6	14.6
	BLUR	85.2	74.5	69.4	40.9	75.8	66.7	54.6	**100.0**	33.3	69.2
	NOISE	**94.8**	70.8	83.1	**21.2**	**81.7**	**96.0**	65.5	**100.0**	**4.0**	**84.1**
	PRUNE	48.4	80.4	88.6	57.1	67.6	73.4	77.4	99.4	27.0	82.0
LLaVA-OV-7B	PROMPT	15.3	**94.3**	**100.0**	84.7	34.8	32.1	**92.8**	**100.0**	67.9	57.8
	BLUR	80.8	74.6	73.1	40.9	76.0	86.1	72.3	**100.0**	13.9	84.6
	NOISE	**94.4**	60.6	83.4	**21.2**	**76.8**	**100.0**	56.5	**100.0**	**0.0**	79.6
	PRUNE	43.2	81.0	99.3	57.1	65.8	88.3	75.7	**100.0**	11.7	**86.9**
Qwen2-VL-7B	PROMPT	22.8	**91.5**	**100.0**	77.2	46.3	63.6	**83.1**	**100.0**	36.4	79.5
	BLUR	84.1	81.1	70.3	40.9	**78.0**	96.4	68.3	**100.0**	3.6	**85.7**
	NOISE	**100.0**	43.9	78.8	**21.2**	66.0	**100.0**	33.2	**100.0**	**0.0**	59.9
	PRUNE	86.3	68.1	49.7	57.1	64.6	**100.0**	65.2	**100.0**	**0.0**	84.9
Qwen2.5-VL-7B	PROMPT	87.1	**86.3**	33.3	71.0	56.5	79.0	**85.3**	**100.0**	21.1	87.2
	BLUR	88.5	79.0	66.8	40.9	**77.1**	98.7	78.2	**100.0**	1.3	**91.1**
	NOISE	**99.7**	50.0	79.0	**21.2**	70.3	**100.0**	48.5	**100.0**	**0.0**	73.9
	PRUNE	28.5	79.2	**100.0**	71.5	52.0	72.4	80.4	**100.0**	27.6	82.7

Post-Training Methods. We consider four strategies (as shown in Fig. 1):

1. `Supervised Fine-Tuning` (*sft*), which maximizes the log-likelihood of preferred responses to enforce positive alignment;
2. `Gradient Ascent` (*ga*) [49], a machine unlearning technique that penalizes undesirable outputs by performing gradient ascent on rejected samples;
3. `Direct Preference Optimization` (*dpo*) [55], which optimizes pairwise preferences between chosen and rejected outputs;
4. `Negative Preference Optimization` (*npo*) [68], a machine unlearning approach penalizing rejected responses and divergence from the original model (on rejected ones) jointly.

Implementation Details. All hyperparameters are validated on TUTUS-D. For training-free methods, we set γ=2 for NOISE and k=0.7 for PRUNE (see Appendix G, especially Appendix G.2, for implementation details and different values). For post-training, we use LoRA [26] with r=16, α=16, and dropout 0.5 (see Appendix G.1 for hyperparameter tuning). For TUTUS-F, we simulate downstream identity memorization by first fine-tuning the language model on fictitious identities using the same LoRA settings, before evaluating anonymization methods (details in Appendix I).

5.2. The Limits of Training-Free Solutions

We benchmark all training-free methods on both TUTUS-C and TUTUS-F. In Fig. 2, we illustrate the trade-off between successful anonymization (*rss*) and content preservation (*cdp*) for all training-free methods across both subsets, while we report numerical results including *jdr* in Tab. 1.

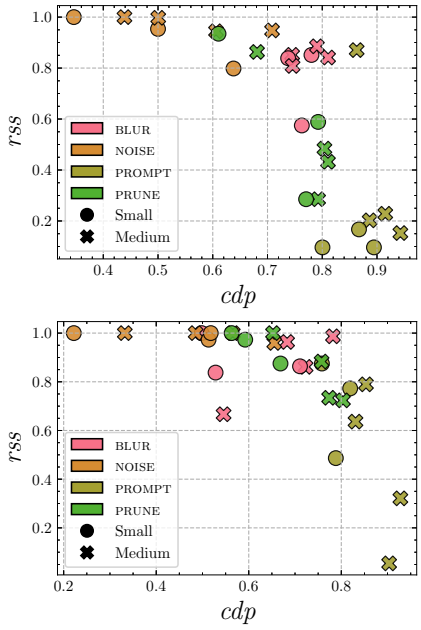


Figure 2. Training-free anonymization on TUTUS-C (top) and TUTUS-F (bottom) for small (circle) and medium (cross) models. Identity suppression (*rss*) versus preservation (*cdp*) reveals a Pareto frontier: high *rss* often degrades content (NOISE), while preservation-oriented methods fail to suppress identity (PROMPT). BLUR and PRUNE offer intermediate trade-offs, but lower *rss* on TUTUS-C suggests pretrained identities are harder to suppress and identity cues extend beyond faces.

Identity Suppression and Preservation. In general, on both subsets, training-free methods form a Pareto frontier, highlighting that achieving near-perfect *rss* and *cdp* jointly is extremely challenging. From Fig. 2, we can see that NOISE is effective in identity suppression at the expense of preservation, excessively disrupting features. Differently, PROMPT achieves lower *rss* but higher *cdp*, highlighting that naive text-based instructions are not a viable approach for protecting identities. In contrast, BLUR and PRUNE display better trade-offs even if their *rss* tends to be lower on TUTUS-C than on TUTUS-F, hinting that erasing pretraining knowledge is more challenging than protecting identities learned at adaptation only. Notably, BLUR saturates at around 0.85 *rss*. This suggests that *identity cues go beyond facial regions*, involving other visual attributes, *e.g.*, hairstyle or clothing. This has notable implications, since BLUR is the *de facto* standard adopted in dataset creation [20] and commercial services [22].

Robustness. From Tab. 1, we can see that stronger instruction-tuned models like Qwen2.5-VL-7B exhibit better compliance with the input query, achieving high *rss*

Table 2. Results of **post-training** methods on TUTUS-C with residual malicious leakage on eligible examples. For each {objective × model}, we report the best-performing component; bold is the best in each group. S and U denote seen (identities in TUTUS-T) and unseen identities, respectively.

Model	Method	Best Component	*rss*↑		*cdp*↑		*jdr*↑		*irm*↓		*hm*↑	
			S	U	S	U	S	U	S	U	S	U
Small Models												
LLaVA-OV-0.5B	*sft*	CONNECTOR	90.5	90.1	70.9	69.1	95.9	93.0	13.2	16.2	84.3	82.6
	ga	LLM	98.9	94.6	**79.8**	**81.2**	94.7	**100.0**	6.4	5.4	**90.4**	**91.2**
	dpo	CONNECTOR	**99.5**	**99.2**	65.0	64.8	**98.9**	98.2	**1.6**	**2.5**	84.4	84.1
	npo	LLM	98.9	94.6	79.5	80.3	94.7	**100.0**	6.4	5.4	90.2	90.9
Qwen2.5-VL-3B	*sft*	VISION	75.2	75.6	73.5	70.3	46.6	40.0	65.0	69.8	62.0	57.2
	ga	VISION	74.3	80.2	**73.8**	**72.9**	74.8	53.6	44.4	57.0	74.3	66.9
	dpo	VISION	97.2	**98.8**	60.4	62.0	**100.0**	**96.5**	**2.8**	**4.6**	**81.4**	**81.9**
	npo	VISION	**99.5**	**98.8**	61.3	62.8	86.4	88.2	14.0	12.8	79.1	80.3
Medium Models												
LLaVANext-7B	*sft*	VISION	97.9	97.2	66.7	67.0	94.7	89.4	7.3	13.1	83.9	82.4
	ga	LLM	99.0	98.1	79.5	**79.6**	88.0	88.6	13.0	13.1	**88.1**	**88.1**
	dpo	VISION	**99.5**	**100.0**	58.1	57.8	**99.5**	**100.0**	**1.0**	**0.0**	80.4	80.4
	npo	LLM	98.5	97.2	**80.0**	79.5	85.8	88.5	15.5	14.0	87.4	87.8
LLaVA-OV-7B	*sft*	VISION	98.5	98.1	84.1	**85.3**	83.5	86.1	17.8	15.5	88.2	89.5
	ga	LLM	97.5	94.2	79.1	79.6	**100.0**	**100.0**	**2.5**	**5.8**	**91.2**	90.4
	dpo	LLM	**99.5**	**99.0**	**84.5**	85.2	80.6	88.2	19.8	12.6	87.5	90.4
	npo	LLM	**99.5**	**99.0**	83.3	83.4	91.8	94.1	8.6	6.8	91.1	**91.7**
Qwen2-VL-7B	*sft*	VISION	**100.0**	**100.0**	62.8	63.3	80.3	78.2	19.7	21.8	78.2	77.8
	ga	VISION	93.4	90.8	62.0	61.9	90.0	87.4	16.0	20.7	79.0	77.7
	dpo	VISION	**100.0**	98.9	57.3	56.5	**100.0**	**100.0**	**0.0**	**1.1**	**80.1**	**79.3**
	npo	VISION	81.7	75.9	**71.8**	**70.2**	71.3	63.6	41.8	51.7	74.6	69.5
Qwen2.5-VL-7B	*sft*	VISION	**93.5**	**90.7**	**67.6**	**70.4**	16.5	9.0	84.6	91.9	34.8	21.9
	ga	CONNECTOR	86.0	86.1	58.3	56.9	**82.6**	**82.4**	**29.0**	**29.1**	**73.4**	**72.6**
	dpo	VISION	83.6	**90.7**	64.4	62.9	59.8	39.7	50.0	64.0	67.9	57.6
	npo	CONNECTOR	85.5	84.9	58.7	57.2	82.0	82.2	29.9	30.2	73.3	72.4

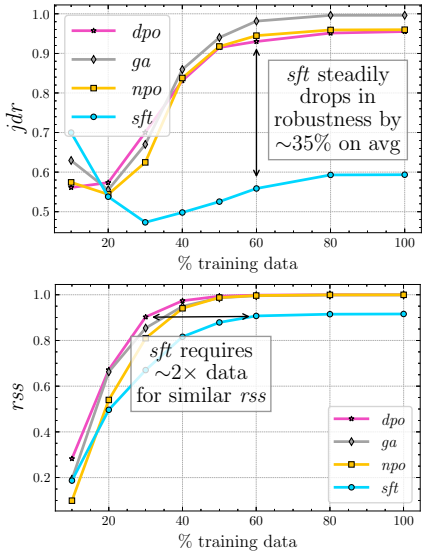


Figure 3. Post-training objective comparison. Robustness (top) and identity suppression vs. training data (bottom). The two plots show a clear disadvantage in choosing *sft* for post-training alignment compared to the other objectives (*dpo*, *npo*, and *sft*).

(87.1%) with the PROMPT baseline. Surprisingly, this capability backfires: since these models accurately follow user prompts, they are also more vulnerable even to simple jailbreak instructions with a considerable low *jdr* (33.3%). Indeed, the corresponding irm of 71.0% shows that the malicious prompt still elicits identity leakage in more than two-thirds of eligible examples. Interestingly, a similar irm (71.5) for PRUNE on the same model, corresponds to a higher *jdr*. This is due to the low *rss* of PRUNE, which makes *jdr* saturate, highlighting the importance of reporting a complementary value as the irm.

5.3. Post-Training Anonymization

Motivated by the limitations of training-free solutions, we study *any identity anonymization* via post-training. Using TUTUS-T as our training set, we evaluate four alignment strategies (Sec. 5.1) across three model components: the vision encoder (VISION), the connector (CONNECTOR), and the language model (LLM). This yields 72 configurations from {model × objective × component} combinations. For clarity, Tab. 2 reports results on `seen` (*i.e.*, identities in TUTUS-T) and `unseen` identities of TUTUS-C, showing each objective with its best component. Appendix H provides the full per-model breakdown.

Models. There are disparities among architectures in terms of both performance and best design choice. For instance, while tuning the vision encoder is overall the optimal choice regardless of the objective, LLaVA-OV-0.5B and LLaVA-OV-7B benefit from post-training applied to LLM. Similarly, CONNECTOR performs well on Qwen2.5-VL-7B under *ga* and *npo*, but is outperformed by VISION in Qwen2-VL-7B. Overall, there is no single design (*i.e.*, no single objective-component combination) consistently outperforming the others across models, indicating that privacy alignment is *strongly model-dependent*.

Post-Training Objectives. Tab. 2 shows no clear winner among objectives. We inspect this trend in Fig. 3, analyzing how *jdr* (left) and *rss* (right) vary with training data, averaging each objective across models and components. In Fig. 3 (left), *sft* is the weakest choice, dropping ≈35% *jdr* on average compared to *dpo*, *npo*, and *ga*, which show stronger robustness. This supports the observation in Sec. 5.2 that better instruction-following, as enforced by *sft*, enables prompt-driven attacks. Moreover, Fig. 3 (right) shows that *dpo* achieves outstanding *rss*, while *sft* consistently plateaus around 90%. Finally, *sft* requires 2× more data than *dpo* to reach comparable peak performance, which is critical when curated preference pairs are limited and hyperparameter selection is challenging.

Components. We further examine the insight of Tab. 2 that VISION is a good choice for anonymization by averaging results of different models and objectives under the same component, reporting them in Fig. 4. From Fig. 4 (left), tuning the *vision encoder* (VISION) consistently outperforms LLM and CONNECTOR, regardless of the training objective, yielding an average +15% in *hm* compared to the second-best. Additionally, adapting the *vision encoder* leads to the

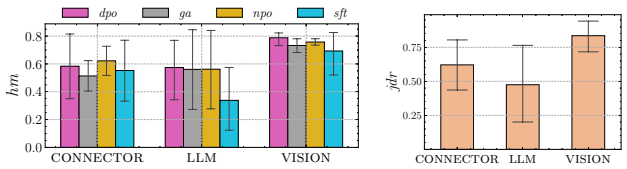


Figure 4. Training performance and robustness comparison across model components. Regardless of the objective, training the VISION towards anonymization yields the highest overall performance (left) and best robustness (right). Conversely, tuning the LLM leads to the worst trade-off.

Table 3. Benign prompt variation. Ref. is refusal-marker rate, Leak is identity leakage used only as a diagnostic, and Sim. is response similarity to the pristine model.

Benign query	Samples	Ref.	Leak	Sim.
Action	548	0.0	0.23	.615
Clothing/accessories	548	3.3	0.00	.696
Scene/background	548	0.9	0.37	.645
Expression/pose	548	4.2	0.00	.654
Visible attributes	548	1.3	0.14	.632
Avg.	548	1.9	0.15	.649

highest robustness by a large margin (Fig. 4, right): $+20\%$ *jdr* w.r.t. CONNECTOR and $+30\%$ *jdr* w.r.t. LLM. These observations favor vision-encoder alignment in the evaluated LLaVA and Qwen models and settings. This is important since LLM+*sft* is arguably the most popular choice for safety alignment [40, 44], yet our results elicit a better practice.

Refusal Analysis. We test whether anonymization comes from blanket refusal and human-related features elimination or not. The benign-prompt analysis on a restricted subset of samples (548) averaged across models on VISION +*dpo* setting in Tab. 3 shows that, across five benign human-centric prompts, the average refusal rate is 1.9%, benign identity leakage is 0.15%, and answer similarity to the pristine model is 0.649. Qualitatively, on one shared evaluation image, the anonymized model avoids identity disclosure with a generic person description, yet still describes the person's action, clothing, and pose (Appendix Tab. 8). Thus, VISION alignment mainly suppresses identity disclosure while preserving non-identifying human semantics.

5.4. Transfer to Unseen Identities

As we aim to suppress identity cues for *any* individual, a core question is whether post-training generalizes beyond `seen` identities (*i.e.*, in our setup, beyond the identities in TUTUS-T). Therefore, in Fig. 5, we report harmonic mean difference (Δhm) between SEEN and UNSEEN individuals.

Results on TUTUS-C. Average seen–unseen *hm* gaps are below four percentage points (Fig. 5), with smaller gaps for *dpo* than *ga* and *npo*. Small gaps indicate limited celebrity-domain degradation, not strong absolute unseen

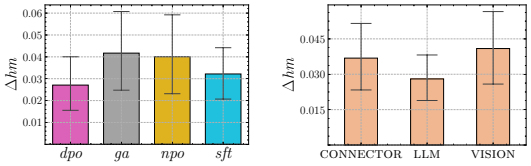


Figure 5. Seen-to-unseen generalization gap (Δhm) across objectives (left) and trainable components (right). Average gaps remain below four percentage points; interpret them alongside the absolute unseen scores in Tab. 2.

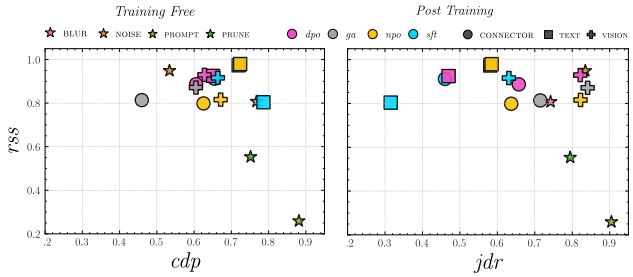


Figure 6. We compare training-free and post-training methods for identity suppression vs preservation (left) and robustness (right). PRUNE and PROMPT show weak *rss* despite high *cdp* and *jdr*, while post-training the *vision encoder* (cross) yields consistently strong *rss*. However, *sft* (azure) underperforms other objectives.

performance; see Tab. 2. Gap ratios do not imply multiplicative performance gains. Even in this scenario, VISION is the best alignment choice, close to CONNECTOR, with LLM resulting in lower generalization.

5.5. Comparison with Training-Free Methods.

We compare the best post-training configurations (in Tab. 2) with training-free baselines (Tab. 1) on the 200 `unseen` identities of TUTUS-C (post-training is exposed to the remaining 300 identities in TUTUS-T, see Sec. 4.1) Across models, post-training consistently outperforms BLUR, the strongest training-free baseline, achieving higher *rss* and *jdr*. For instance, LLaVA-OV-7B (LLM +*dpo*) reaches 99.0% *rss* compared to 80.8% for BLUR, alongside higher robustness (88.2% *jdr* vs 73.1%), while maintaining better preservation (85.2% *cdp* vs 74.6%). Similarly, with Qwen2-VL-7B, VISION+*dpo* improves *rss* by about $+15\%$ and further increases *jdr*, at the cost of a modest drop in *cdp* compared to BLUR, which only modifies facial regions. Compared to NOISE, which achieves the highest *rss* among training-free methods, post-training maintains similar suppression while improving both *cdp* and *jdr*. Figure 6 summarizes these trends. Training-free approaches (*e.g.*, PROMPT and PRUNE) achieve higher *cdp* or *jdr* but consistently lower *rss*. In contrast, post-training methods achieve strong *rss* with competitive *cdp* and *jdr*, particularly when applied to the VISION encoder (crosses), which dom-

Table 4. **Post-training** results on TUTUS-F with residual malicious leakage on eligible examples. Components are selected as in the TUTUS-C table.

Model	Method	Best Com.	*rss*↑	*cdp*↑	*jdr*↑	irm↓	*hm*↑
Small Models							
LLaVA-OV-0.5B	*sft*	CONNECTOR	**100.0**	54.9	**100.0**	**0.0**	78.5
	ga	LLM	43.2	78.9	**100.0**	56.8	65.5
	dpo	CONNECTOR	**100.0**	56.6	**100.0**	**0.0**	**79.7**
	npo	LLM	40.5	**79.7**	**100.0**	59.5	63.5
Qwen2.5-VL-3B	*sft*	VISION	**100.0**	**60.8**	**100.0**	**0.0**	**82.3**
	ga	VISION	**100.0**	59.9	**100.0**	**0.0**	81.7
	dpo	VISION	**100.0**	51.1	**100.0**	**0.0**	75.8
	npo	VISION	**100.0**	35.5	**100.0**	**0.0**	62.3
Medium Models							
LLaVANext-7B	*sft*	VISION	94.1	69.6	**100.0**	5.9	**85.8**
	ga	LLM	16.7	77.8	**100.0**	83.3	36.2
	dpo	VISION	**99.8**	51.9	99.6	**0.7**	76.2
	npo	LLM	15.3	**77.8**	**100.0**	84.7	34.0
LLaVA-OV-7B	*sft*	VISION	83.2	81.5	**100.0**	16.8	87.5
	ga	LLM	**100.0**	83.9	98.5	**1.5**	**93.6**
	dpo	LLM	92.0	**87.2**	**100.0**	8.0	92.8
	npo	LLM	88.3	84.3	**100.0**	11.7	90.4
Qwen2-VL-7B	*sft*	VISION	**100.0**	53.6	**100.0**	**0.0**	77.6
	ga	VISION	**100.0**	56.7	**100.0**	**0.0**	79.7
	dpo	VISION	**100.0**	49.5	**100.0**	**0.0**	74.7
	npo	VISION	**100.0**	**67.2**	**100.0**	**0.0**	**86.0**
Qwen2.5-VL-7B	*sft*	VISION	**100.0**	**63.5**	98.7	1.3	**83.6**
	ga	CONNECTOR	**100.0**	61.8	**100.0**	**0.0**	82.9
	dpo	VISION	98.7	62.6	**100.0**	1.3	83.1
	npo	CONNECTOR	**100.0**	62.7	**100.0**	**0.0**	83.5

inates the Pareto frontier confirming the benefits of visual-level alignment observed in previous sections. Among objectives, *sft* remains weaker than alternatives. To verify that VISION tuning is not tuned only to the default malicious suffix, we further stress-test VISION+*dpo* with stronger prompt attacks on pristine-leaking samples (Appendix D). Across this attack set, the forced name-only request is consistently the strongest attack; even then, average identity leakage remains 6.8% (26.4% in the weakest model), and average *jdr* remains 94.4%. In Appendix F we report results on established MLLMs benchmarks [19, 34, 43, 66, 71], confirming these *cdp* trends.

In conclusion, post-training *any identity anonymization* outperforms training-free alternatives, though better trade-offs remain open. We next stress-test post-training under continual adaptation.

6. Adaptation to Unknown Identities

So far, we evaluated anonymization in a static setting, where a model undergoes anonymization *after training*. We now test *continual adaptation*: whether anonymization adapters trained before personalization can still protect new identities learned later through downstream fine-tuning.

Implementation Details. Following the fictitious-identity protocol of Sec. 5.1, we adapt the language model to TUTUS-F identities via LoRA [26] (r=16, α=16, dropout 0.5; see Appendix I). We then linearly combine [65] the LoRA adapter trained on TUTUS-T with the updated model, without further training, and evaluate anonymization transfer to new identities using *rss*, *cdp*, and *jdr*.

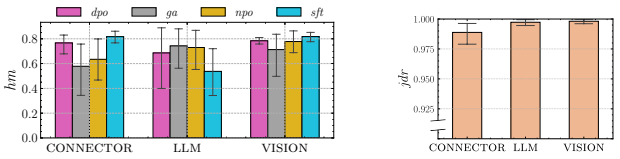


Figure 7. Training the *vision encoder* yields the best anonymization performance, while *language model* tuning to the worst trade-off (left). Robustness differences across components remain small, likely because base model identity recognition in this setting is lower than for pretraining knowledge (see Appendix B).

Results. Table 4 and Fig. 7 show that VISION again transfers best (+5% average *hm*), suggesting that visual anonymization interferes less with downstream text-decoder adaptation. Even text-side LoRA adapters retain non-trivial *rss* and *cdp*, while non-zero *cdp* indicates that adapters generalize beyond simply destroying image representations. Overall, anonymization persists after fine-tuning on new identities without retraining.

7. Conclusion

We introduced *any identity anonymization*, the problem of preventing identity leakage for *any* individual in MLLMs beyond predefined forget sets. Our benchmark, TUTUS, jointly evaluates identity suppression, robustness to prompt attacks, and preservation of model utility. Our results show that simple training-free methods do not reliably prevent identity disclosure, while post-training alignment achieves better trade-offs among the three desiderata. Within the evaluated families and settings, vision-encoder alignment generally offers favorable trade-offs. Post-training anonymization remains effective in continual adaptation scenarios, where new identities are introduced, without requiring additional fine-tuning. TUTUS exposes three gaps: prompt vulnerability despite high direct suppression, loss of descriptive utility, and untested protection for arbitrary individuals. These motivate stronger attacks, human-validated utility, and broader identity coverage.

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References

[1] Martin Abadi, Andy Chu, Ian Goodfellow, H Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. Deep learning with differential privacy. In *ACM SIGSAC conference on computer and communications security*, 2016. 2

[2] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv:2303.08774*, 2023. 3

[3] Yuval Alaluf, Elad Richardson, Sergey Tulyakov, Kfir Aberman, and Daniel Cohen-Or. Myvlm: Personalizing vlms for user-specific queries. In *European Conference on Computer Vision*, pages 73–91. Springer, 2024. 3

[4] Saketh Bachu, Erfan Shayegani, Rohit Lal, Trishna Chakraborty, Arindam Dutta, Chengyu Song, Yue Dong, Nael B Abu-Ghazaleh, and Amit Roy-Chowdhury. Layer-wise alignment: Examining safety alignment across image encoder layers in vision language models. In *ICML*, 2025. 2

[5] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, et al. Qwen2.5-vl technical report. 2025. 1, 2, 4, 12

[6] Simone Barattin, Christos Tzelepis, Ioannis Patras, and Nicu Sebe. Attribute-preserving face dataset anonymization via latent code optimization. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 8001–8010, 2023. 18

[7] Alberto Blanco-Justicia, David Sánchez, Josep Domingo-Ferrer, and Krishnamurty Muralidhar. A critical review on the use (and misuse) of differential privacy in machine learning. *ACM Computing Surveys*, 55(8):1–16, 2022. 2

[8] Lucas Bourtoule, Varun Chandrasekaran, Christopher A Choquette-Choo, Hengrui Jia, Adelin Travers, Baiwu Zhang, David Lie, and Nicolas Papernot. Machine unlearning. In *IEEE Symposium on Security and Privacy (SP)*, 2021. 2

[9] Simone Caldarella, Massimiliano Mancini, Elisa Ricci, and Rahaf Aljundi. The phantom menace: Unmasking privacy leakages in vision-language models. *ECCV-WS*, 2024. 2, 3, 14, 15

[10] Nicholas Carlini, Steve Chien, Milad Nasr, Shuang Song, Andreas Terzis, and Florian Tramer. Membership inference attacks from first principles. In *2022 IEEE SP*, 2022. 2

[11] Nicola DallAsen, Yiming Wang, Hao Tang, Luca Zanella, and Elisa Ricci. Graph-based generative face anonymisation with pose preservation. In *ICIAP*, 2022. 18

[12] Thomas De Min, Massimiliano Mancini, Stéphane Lathuilière, Subhankar Roy, and Elisa Ricci. Unlearning personal data from a single image. *TMLR*, 2025. 2

[13] Arnab Dhar. Yolov8 face detection. Hugging Face, 2023. 4

[14] Yi Ding, Bolian Li, and Ruqi Zhang. Eta: Evaluating then aligning safety of vision language models at inference time. *ICLR*, 2025. 2

[15] Cynthia Dwork. Differential privacy. In *International colloquium on automata, languages, and programming*, pages 1–12. Springer, 2006. 2

[16] Chongyu Fan, Jiancheng Liu, Yihua Zhang, Eric Wong, Dennis Wei, and Sijia Liu. Salun: Empowering machine unlearning via gradient-based weight saliency in both image classification and generation. *ICLR*, 2024. 2

[17] Chongyu Fan, Jiancheng Liu, Licong Lin, Jinghan Jia, Ruiqi Zhang, Song Mei, and Sijia Liu. Simplicity prevails: Rethinking negative preference optimization for llm unlearning. 2025. 2

[18] Jack Foster, Stefan Schoepf, and Alexandra Brintrup. Fast machine unlearning without retraining through selective synaptic dampening. In *AAAI*, 2024. 2

[19] Chaoyou Fu, Peixian Chen, Yunhang Shen, Yulei Qin, Mengdan Zhang, Xu Lin, Jinrui Yang, Xiawu Zheng, Ke Li, Xing Sun, et al. Mme: A comprehensive evaluation benchmark for multimodal large language models. *arxiv*, 2023. 8, 15

[20] Samir Yitzhak Gadre, Gabriel Ilharco, Alex Fang, Jonathan Hayase, Georgios Smyrnis, Thao Nguyen, Ryan Marten, Mitchell Wortsman, Dhruba Ghosh, Jieyu Zhang, et al. Datacomp: In search of the next generation of multimodal datasets. *NeurIPS*, 2023. 2, 5

[21] Oran Gafni, Lior Wolf, and Yaniv Taigman. Live face de-identification in video. In *ICCV*, 2019. 2

[22] Google. Google maps, . Accessed: 2026-3-3. 5

[23] Google. Google trends, . Accessed: 2026-3-3. 12, 13

[24] Ralph Gross, Latanya Sweeney, Fernando De la Torre, and Simon Baker. Model-based face de-identification. In *CVPR-WS*, 2006. 2

[25] Dominik Hintersdorf, Lukas Struppek, Manuel Brack, Felix Friedrich, Patrick Schramowski, and Kristian Kersting. Does clip know my face? *Journal of Artificial Intelligence Research*, 2024. 2

[26] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, et al. Lora: Low-rank adaptation of large language models. *ICLR*, 2022. 2, 5, 8

[27] Håkon Hukkelås, Rudolf Mester, and Frank Lindseth. Deepprivacy: A generative adversarial network for face anonymization. In *International symposium on visual computing*, 2019. 2

[28] Bargav Jayaraman, Chuan Guo, and Kamalika Chaudhuri. Déjà vu memorization in vision–language models. *NeurIPS*, 2024. 2

[29] Tianjie Ju, Yi Hua, Hao Fei, Zhenyu Shao, Yubin Zheng, Haodong Zhao, Mong-Li Lee, Wynne Hsu, Zhuosheng Zhang, and Gongshen Liu. Watch out your album! on the inadvertent privacy memorization in multi-modal large language models. In *ICML*, 2025. 2

[30] Tero Karras, Timo Aila, Samuli Laine, and Jaakko Lehtinen. Progressive growing of gans for improved quality, stability, and variation. In *ICLR*, 2018. 15

[31] Han-Wei Kung, Tuomas Varanka, Sanjay Saha, Terence Sim, and Nicu Sebe. Face anonymization made simple. In *WACV*, 2025. 2, 18

[32] Meghdad Kurmanji, Peter Triantafillou, Jamie Hayes, and Eleni Triantafillou. Towards unbounded machine unlearning. *NeurIPS*, 2024. 2

[33] Quentin Lhoest, Albert Villanova del Moral, Yacine Jernite, et al. The HuggingFace hub: A central repository for machine learning models. In *EMNLP*, 2021. 16

[34] Bohao Li, Rui Wang, Guangzhi Wang, Yuying Ge, Yixiao Ge, and Ying Shan. Seed-bench: Benchmarking multimodal llms with generative comprehension. *CVPR*, 2023. 8, 15

[35] Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Peiyuan Zhang, Yanwei Li, Ziwei Liu, et al. Llava-onevision: Easy visual task transfer. 2024. 1, 2, 4, 12

[36] Mukai Li, Lei Li, Yuwei Yin, Masood Ahmed, Zhenguang Liu, and Qi Liu. Red teaming visual language models. *arXiv preprint arXiv:2401.12915*, 2024. 2

[37] Shen Li, Yanli Zhao, Rohan Varma, Omkar Salpekar, Pieter Noordhuis, Teng Li, Adam Paszke, Jeff Smith, Brian Vaughan, Pritam Damania, et al. Pytorch distributed: Experiences on accelerating data parallel training. *arXiv preprint arXiv:2006.15704*, 2020. 16

[38] Tao Li and Lei Lin. Anonymousnet: Natural face de-identification with measurable privacy. In *CVPR-WS*, 2019. 2

[39] Minxing Liu, Minh-Ha Le, and Niklas Carlsson. From privacy chains to chainshield: Structured privacy risks and defense in vision-language models. *WPES@ACM*, 2025. 2

[40] Qin Liu, Chao Shang, Ling Liu, Nikolaos Pappas, Jie Ma, Neha Anna John, Srikanth Doss, Lluis Marquez, Miguel Ballesteros, and Yassine Benajiba. Unraveling and mitigating safety alignment degradation of vision-language models. In *ACL*, 2025. 2, 3, 7

[41] Xiaogeng Liu, Nan Xu, Muhao Chen, and Chaowei Xiao. Autodan: Generating stealthy jailbreak prompts on aligned large language models. *arXiv preprint arXiv:2310.04451*, 2023. 3

[42] Xin Liu, Yichen Zhu, Jindong Gu, Yunshi Lan, Chao Yang, and Yu Qiao. Mm-safetybench: A benchmark for safety evaluation of multimodal large language models. In *ECCV*, 2024. 2

[43] Yuan Liu, Haodong Duan, Yuanhan Zhang, Bo Li, Songyang Zhang, Wangbo Zhao, Yike Yuan, Jiaqi Wang, Conghui He, Ziwei Liu, et al. Mmbench: Is your multi-modal model an all-around player? In *ECCV*, 2024. 8, 15

[44] Zhendong Liu, Yuanbi Nie, Yingshui Tan, Xiangyu Yue, Qiushi Cui, Chongjun Wang, Xiaoyong Zhu, and Bo Zheng. Safety alignment for vision language models. *arXiv preprint arXiv:2405.13581*, 2024. 2, 3, 7

[45] Zheyuan Liu, Guangyao Dou, Mengzhao Jia, Zhaoxuan Tan, Qingkai Zeng, Yongle Yuan, and Meng Jiang. Protecting privacy in multimodal large language models with mllmu-bench. *NAACL*, 2025. 3

[46] Zheyuan Liu, Guangyao Dou, Mengzhao Jia, Zhaoxuan Tan, Qingkai Zeng, Yongle Yuan, and Meng Jiang. Protecting privacy in multimodal large language models with mllmu-bench. In *NAACL*, 2025. 2, 15, 17

[47] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017. 17

[48] Yingzi Ma, Jiongxiao Wang, Fei Wang, Siyuan Ma, Jiazhao Li, Jinsheng Pan, Xiujun Li, Furong Huang, Lichao Sun, Bo Li, et al. Benchmarking vision language model unlearning via fictitious facial identity dataset. *ICLR*, 2025. 2

[49] Pratyush Maini, Zhili Feng, Avi Schwarzschild, Zachary C Lipton, and J Zico Kolter. Tofu: A task of fictitious unlearning for llms. *COLM*, 2024. 2, 5

[50] Meta Platforms, Inc. Meta ray-ban display: Breakthrough ai glasses available now, 2025. Accessed: 2026-3-3. 12, 13

[51] Yeongtak Oh, Dohyun Chung, Juhyeon Shin, Sangha Park, Johan Barthelemy, Jisoo Mok, and Sungroh Yoon. Repic: Reinforced post-training for personalizing multi-modal language models. *NeurIPS*, 2025. 3

[52] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. Pytorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*, 2019. 16

[53] Youyang Qu, Xin Yuan, Ming Ding, Wei Ni, Thierry Rakotoarivelo, and David Smith. Learn to unlearn: A survey on machine unlearning. *arXiv*, 2023. 2

[54] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *ICML*, 2021. 2

[55] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *NeurIPS*, 2023. 2, 3, 5, 17

[56] Nils Reimers and Iryna Gurevych. all-minilm-l6-v2: A sentence-transformer model, 2021. Accessed: 2025-03-08. 4

[57] Jie Ren, Yaxin Li, Shenglai Zeng, Han Xu, Lingjuan Lyu, Yue Xing, and Jiliang Tang. Unveiling and mitigating memorization in text-to-image diffusion models through cross attention. In *ECCV*, 2024. 2

[58] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *MICCAI*, 2015. 18

[59] Fahad Shamshad, Muzammal Naseer, and Karthik Nandakumar. Clip2protect: Protecting facial privacy using text-guided makeup via adversarial latent search. In *CVPR*, 2023. 2

[60] Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. Membership inference attacks against machine learning models. In *2017 IEEE symposium on security and privacy (SP)*, 2017. 2

[61] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, et al. Qwen2-vl: Enhancing vision-language model's perception of the world at any resolution. *arXiv*, 2024. 4

[62] Ryan Webster and Teddy Furon. Multi-modal identity extraction. In *ICCV*, 2025. 2

[63] Alexander Wei, Nika Haghtalab, and Jacob Steinhardt. Jailbroken: How does llm safety training fail? *Advances in neural information processing systems*, 36:80079–80110, 2023. 3

[64] Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim

Rault, Remi Louf, Morgan Funtowicz, and Jamie Brew. Transformers: State-of-the-art natural language processing. In *EMNLP*, 2020. 16

[65] Mitchell Wortsman, Gabriel Ilharco, Samir Ya Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, et al. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *ICML*, 2022. 8

[66] Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, et al. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In *CVPR*, 2024. 8, 15

[67] Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. Sigmoid loss for language image pre-training. In *ICCV*, 2023. 2

[68] Ruiqi Zhang, Licong Lin, Yu Bai, and Song Mei. Negative preference optimization: From catastrophic collapse to effective unlearning. *COLM*, 2024. 2, 5, 17

[69] Yongting Zhang, Lu Chen, Guodong Zheng, Yifeng Gao, Rui Zheng, Jinlan Fu, Zhenfei Yin, Senjie Jin, Yu Qiao, Xuanjing Huang, et al. Spa-vl: A comprehensive safety preference alignment dataset for vision language models. In *CVPR*, 2025. 2

[70] Yitong Zhang, Jia Li, Liyi Cai, and Ge Li. Davsp: Safety alignment for large vision-language models via deep aligned visual safety prompt. *arXiv:2506.09353*, 2025. 2

[71] Yuhui Zhang, Yuchang Su, Yiming Liu, Xiaohan Wang, James Burgess, Elaine Sui, Chenyu Wang, Josiah Aklilu, Alejandro Lozano, Anjiang Wei, et al. Automated generation of challenging multiple-choice questions for vision language model evaluation. In *CVPR*, 2025. 4, 8, 13, 15

[72] Andy Zou, Zifan Wang, Nicholas Carlini, Milad Nasr, J Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*, 2023. 3